Uniform Turán densities of small stars

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Abstract

Let S_k be the 3-uniform hypergraph whose edges consist of a fixed centre and any two of k leaves. We determine the uniform Turán densities of S_6, S_7, and S_8 as 13/25, 7/12, and 31/49, respectively. Together with the theorem of Lin and Zhou for $k \geq 9$, this proves that the construction of Reiher, Rödl, and Schacht is optimal for every $k \geq 6$. The proof optimizes the colour weights of an arbitrary palette and retains the combined contribution of its three coordinate fibres. For S_6, an intersection estimate and a correction with zero weighted mean provide an additional constraint. The remaining inequalities have exact integer Bernstein certificates. A separate argument using independently sampled colours gives the upper bounds $\pi_{\mathrm{u}}(S_4) \leq 37/100$ and $\pi_{\mathrm{u}}(S_5) \leq 12/25$. All computer-assisted steps are supplied with portable exact verifiers.

1 Introduction

A 3-uniform hypergraph, or 3-graph, is a pair $H = (V, E)$ with $E \subseteq \binom{V}{3}$. For $X, Y, Z \subseteq V$, let $e_H(X, Y, Z)$ count the ordered triples $(x, y, z) \in X \times Y \times Z$ with $\{x, y, z\} \in E$. Given $d \in [0, 1]$ and $\eta > 0$, call H (d, η)-*uniformly dense* if

$$e_H(X, Y, Z) \geq d|X||Y||Z| - \eta|V|^3 \qquad \text{for all } X, Y, Z \subseteq V.$$

For a fixed 3-graph F, its *uniform Turán density* $\pi_{\mathrm{u}}(F)$ is the supremum of the values d such that, for every $\eta > 0$ and every integer n_0, there is an F-free (d, η)-uniformly dense 3-graph on at least n_0 vertices. Copies of F are ordinary subgraphs; they need not be induced.

The k-star S_k has a centre v and k leaves, with all triples containing v and two distinct leaves as edges. Determining its uniform Turán density is a problem highlighted by Schacht [5, Problem 3.4, p. 4650]. The construction of Reiher, Rödl, and Schacht [4, Section 5.3.1] gives

$$\pi_{\mathrm{u}}(S_k) \geq \frac{k^2 - 5k + 7}{(k-1)^2} \qquad (k \geq 3). \tag{1}$$

We recall this construction and its short verification in Section 2. Lamaison and Wu [2] proved equality in (1) for $k \geq 48$. Lin and Zhou [3, Theorem 1.2] established equality for $k \geq 9$ and conjectured it for the remaining values $4 \leq k \leq 8$ [3, Conjecture 5.1]. Our main result settles three of those cases.

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Theorem 1.1. *The uniform Turán densities of the stars with six, seven, and eight leaves are*

$$\pi_{\mathrm{u}}(S_6) = \frac{13}{25}, \qquad \pi_{\mathrm{u}}(S_7) = \frac{7}{12}, \qquad \pi_{\mathrm{u}}(S_8) = \frac{31}{49}.$$

Consequently, $\pi_{\mathrm{u}}(S_k) = (k^2 - 5k + 7)/(k-1)^2$ *for every* $k \geq 6$.

We also obtain bounds for the two smaller unresolved stars, improving the upper bounds 4/9 and 9/16 in [4, Equation (5.3)].

Theorem 1.2. *The uniform Turán densities of* S_4 *and* S_5 *satisfy*

$$\frac{1}{3} \leq \pi_{\mathrm{u}}(S_4) \leq \frac{37}{100}, \qquad \frac{7}{16} \leq \pi_{\mathrm{u}}(S_5) \leq \frac{12}{25}.$$

The lower bounds in both theorems are the known construction in (1). Our contribution is to prove the upper bounds without a restriction on the number of colours. We do not determine $\pi_{\mathrm{u}}(S_4)$ or $\pi_{\mathrm{u}}(S_5)$.

The palette characterization of Lamaison [1] reduces the problem to finite sets of admissible ordered colour triples. We use the auxiliary digraph of Lamaison and Wu [2], together with their reciprocal degree estimate. Lin and Zhou [3, Section 5] explain why their local majorant fails already for S_8 and identify the need for further information about palettes.

We obtain that information by maximizing the density over the colour weights of a fixed palette. At an optimum, the sum of the three coordinate fibres is the same at every positive-weight colour. Keeping this sum intact gives a six-variable constraint that is strong enough for S_7 and S_8, even when an individual fibre is small. For S_6, the overlap of two projection neighbourhoods forces additional missing arc mass. A linear correction whose weighted mean is zero then makes the local estimate compatible with the reciprocal degree bounds.

These local inequalities are proved by finite partitions with exact integer Bernstein certificates and an analytic argument near equality. The smaller-star estimates use a different certificate. Three independent colours determine a finite relation pattern; positive semidefinite forms and weighted neighbourhood inequalities bound its expected density. Repeated sampled colours are retained, so this proof has no asymptotic sampling error. The accompanying supplement contains the rational data and verifiers for both arguments. Section 2 supplies the common reduction, Section 3 proves Theorem 1.1, and Section 4 proves Theorem 1.2.

2 Palettes and projection digraphs

This section reduces the density problem to weighted palettes and records the degree inequalities used in both proofs. We include the auxiliary-digraph argument to specify the directions of its arcs, since the first and third coordinate fibres use different projections.

A *palette* is a set $\mathcal{P} \subseteq \mathcal{C}^3$ for a finite colour set $\mathcal{C}$. For a linearly ordered 3-graph, a colouring of its pairs is *admissible* if every edge $u < v < w$ satisfies

$$(\mathrm{col}(uv), \mathrm{col}(uw), \mathrm{col}(vw)) \in \mathcal{P}.$$

The palette is *F-bad* if no ordering and pair-colouring of F is admissible. For a probability vector $w = (w_i)_{i \in \mathcal{C}}$, define its weighted density by

$$P(w) = \sum_{(i,j,\ell) \in \mathcal{P}} w_i w_j w_\ell.$$

Repeated coordinates are permitted. The palette theorem [1, Theorem 1.1], in the form used in [3, Theorem 2.2], states that

$$\pi_{\mathrm{u}}(F) = \sup\{P(w) : \mathcal{P} \text{ is a finite } F\text{-bad palette},\ w \text{ is a probability vector}\}. \tag{2}$$

The unweighted and weighted formulations have the same supremum. For rational weights, replace each colour by the appropriate number of copies and pull back the admissible triples. A colouring by the copies would project to an admissible original colouring, so badness is preserved. Continuity gives the assertion for real weights.

The following construction of Reiher, Rödl, and Schacht [4, Section 5.3.1] attains every lower bound used here.

Lemma 2.1. *For $r \geq 2$, the palette on $\mathbb{Z}_r$ given by*

$$\mathcal{P}_r = \{(a, b, c) : a \neq b,\ b \neq c,\ c \neq a + 1\}$$

is S_{r+1}-bad and has uniform density $1 - 3/r + 3/r^2$.

Proof. Fixing a, the choice $b = a + 1$ permits $r - 1$ values of c; each of the other $r - 2$ permitted choices of b permits $r - 2$ values. Thus

$$|\mathcal{P}_r| = r\big((r-1) + (r-2)^2\big).$$

Suppose an admissible star colouring exists. For every leaf to the left of the centre, increase its centre-pair colour by one; keep the centre-pair colours of the right leaves unchanged. Among these $r + 1$ labels, two agree. If both leaves are on the left, their star edge has equal second and third colours. If both are on the right, it has equal first and second colours. If they lie on opposite sides, its third colour equals its first colour plus one. Each possibility violates a defining condition of $\mathcal{P}_r$. This proves badness, and division by r^3 gives the density. □

For any palette, write

$$A = \mathrm{proj}_{12}\,\mathcal{P}, \qquad B = \mathrm{proj}_{23}\,\mathcal{P}, \qquad R = \mathrm{proj}_{13}\,\mathcal{P}.$$

We use the same letters for the corresponding 0–1 adjacency functions. Let D have vertex set $\{\mathsf{L}, \mathsf{R}\} \times \mathcal{C}$. Its left side carries B, its right side carries A, and it contains both arcs between (L, i) and (R, j) precisely when $R_{ij} = 1$. In this use, the sans-serif symbols L, R denote sides; the italic R denotes a relation. All digraphs below may have opposite arcs, and a forbidden tournament is an ordinary subgraph. Write T_k for the transitive tournament on k vertices.

Lemma 2.2 (Lamaison–Wu [2])**.** *If $\mathcal{P}$ is S_k-bad, then A and B have no loops and D is T_k-free.*

Proof. A loop in A gives an admissible star by placing the centre first and assigning the loop colour to every centre-pair. A witness for the projection can be chosen independently for each leaf-pair. A loop in B gives the same contradiction with the centre last.

If D contains T_k, retain the tournament order within each side and place all left-side vertices before all right-side vertices. Every required cross-arc remains present because cross-arcs are bidirectional. Place the star centre between these two groups and assign to each centre-pair the colour of its corresponding vertex. The left-left, right-right, and left-right leaf-pairs satisfy B, A, and R, respectively. Choose a witness colour for each leaf-pair. Distinct leaf-pairs correspond to distinct pairs of star vertices, so these choices do not conflict. The resulting star colouring is admissible, a contradiction. □

We use two weighted inequalities for tournament-free digraphs. The first also applies to the neighbourhoods in Section 4.

Lemma 2.3. *Let $k \geq 2$, and let J be a loopless T_k-free digraph with nonnegative vertex masses (w_i) of sum z. Then*

$$\sum_{ij \in E(J)} w_i w_j \leq \frac{k-2}{k-1} z^2. \tag{3}$$

If $z = 1$ and $\mu_i = \max\{\sum_{ij \in E(J)} w_j, \sum_{ji \in E(J)} w_j\}$, then

$$\sum_i \frac{w_i}{1-\mu_i} \leq k-1, \tag{4}$$

where zero-mass vertices are discarded.

Proof. For (3), randomly order the vertices and retain the forward arcs. The resulting underlying graph is K_k-free, since a clique would be a transitive tournament in J. Its weighted edge sum is at most $(k-2)z^2/(2(k-1))$ by the weighted form of Turán's theorem. Every directed arc is retained with probability $1/2$, which proves (3). The weighted graph inequality follows from the finite theorem by rational blow-ups and continuity.

For (4), first suppose that every mass is rational. Replace vertex i by Nw_i independent copies for a common denominator N. A tournament in this blow-up uses at most one vertex from each class, so the blow-up is T_k-free. Each copy of i has maximum normalized in- or outdegree μ_i. The reciprocal degree bound of Lamaison and Wu [2, Lemma 6] gives

$$\sum_i \frac{Nw_i}{1-\mu_i} \leq (k-1)N.$$

Divide by N. For positive real weights, approximate within the same support by rational probability vectors. Since J has no loops, $\mu_i \leq 1 - w_i < 1$, and continuity proves the result. □

For the rest of this section, let $r \geq 2$ and fix a positive probability vector on an S_{r+1}-bad palette. Define the six projection degrees of colour i by

$$\begin{aligned} a_i &= \sum_j A_{ij} w_j, \quad b_i = \sum_j A_{ji} w_j, \quad c_i = \sum_j B_{ij} w_j, \\ d_i &= \sum_j B_{ji} w_j, \quad e_i = \sum_j R_{ij} w_j, \quad f_i = \sum_j R_{ji} w_j. \end{aligned} \tag{5}$$

Set $x_i = \max(a_i, b_i)$ and $y_i = \max(c_i, d_i)$. Applying Lemma 2.3 to A, B, and D gives

$$\sum_i \frac{w_i}{1-x_i} \leq r, \qquad \sum_i \frac{w_i}{1-y_i} \leq r, \tag{6}$$

$$\sum_i w_i \left(\frac{1}{2-y_i-e_i} + \frac{1}{2-x_i-f_i} \right) \leq r. \tag{7}$$

For the last inequality, assign mass $w_i/2$ to each copy of colour i in D. The maximum degrees of its left and right copies are $(y_i+e_i)/2$ and $(x_i+f_i)/2$, respectively. This accounts for both denominators and the factor $1/2$ from each side. The denominators are positive because $x_i, y_i \leq 1 - w_i < 1$.

A final estimate converts palette density into the projection degrees. It is the weighted inclusion-exclusion inequality underlying [3, Lemma 3.1].

Lemma 2.4. *With the notation in* (5)*,*

$$P(w) \le 1 + \sum_i w_i \left(a_i e_i + b_i c_i + d_i f_i - \frac{a_i + b_i + c_i + d_i + e_i + f_i}{2} \right). \tag{8}$$

Proof. Take independent colours I, J, K with law w. Membership of (I, J, K) in $\mathcal{P}$ requires all three events $A_{IJ} = 1$, $B_{JK} = 1$, and $R_{IK} = 1$. Apply inclusion-exclusion to the complementary events and omit the negative triple-intersection term. Their pairwise intersection probabilities are

$$\sum_i w_i(1 - a_i)(1 - e_i), \quad \sum_i w_i(1 - b_i)(1 - c_i), \quad \sum_i w_i(1 - d_i)(1 - f_i).$$

Each relation has the same weighted average indegree and outdegree. Expanding the displayed expressions and using these three equalities gives (8). □

3 The exact values for six, seven, and eight leaves

The reciprocal estimates control an average over colours. To obtain compatible local inequalities, we optimize the weights on each fixed finite palette that would violate the proposed bound. This produces a common constraint on the sum of its three coordinate fibres at every remaining colour. The resulting domain has six variables regardless of the size of the palette; no bound on the number of colours is imposed.

Throughout this section, $r \in \{5, 6, 7\}$ and

$$p_r = 1 - \frac{3}{r} + \frac{3}{r^2}, \qquad q_r = 1 - \frac{1}{r}, \qquad \alpha_r = \frac{r - 2}{2r^3}, \qquad \gamma_r = \frac{3(2r - 3)}{r^2}.$$

Let $\kappa_5 = 1/10$ and $\kappa_6 = \kappa_7 = 0$. For $s, t \in [0, 1]$, put

$$K_r(s, t) = \begin{cases} st - (s + t - 1)_+^2/5, & r = 5, \\ st, & r = 6, 7, \end{cases} \qquad z_+ = \max(z, 0).$$

The subtraction for $r = 5$ measures the missing arcs forced inside an intersection of projection neighbourhoods.

Lemma 3.1. *Suppose an S_{r+1}-bad palette has some weighted density greater than p_r. It has a positive-support restriction and a maximizing probability vector w with value $P^* > p_r$ such that, for every remaining colour,*

$$\min\{q_r, K_r(a_i, e_i)\} + b_i c_i + \min\{q_r, K_r(d_i, f_i)\} \ge 3P^* > 3p_r. \tag{9}$$

Proof. Maximize the homogeneous cubic $P(w)$ on the compact probability simplex, remove zero-weight colours, restrict the palette, and recompute its projections. The value and badness are unchanged. The resulting weight vector is an interior maximizer on its support. If $M_j(i)$ is the total weight of the fibre obtained by fixing coordinate j to i, the Lagrange multiplier equations and Euler's identity give

$$M_1(i) + M_2(i) + M_3(i) = \frac{\partial P}{\partial w_i} = 3P^*.$$

A triple containing colour i more than once contributes once for each occurrence, as required by differentiation.

The first fibre lies among arcs of B from $N_A^+(i)$ to $N_R^+(i)$; these sets have masses a_i and e_i. Therefore $M_1(i) \le a_i e_i$, and also $M_1(i) \le q_r$ by (3) for B. The second fibre has $M_2(i) \le b_i c_i$. The third lies among arcs of A from $N_R^-(i)$ to $N_B^-(i)$, giving $M_3(i) \le \min(q_r, d_i f_i)$. This proves (9) when $r = 6, 7$.

When $r = 5$, the intersection $X = N_A^+(i) \cap N_R^+(i)$ has mass $h \ge (a_i + e_i - 1)_+$. Its induced digraph in B has arc mass at most $4h^2/5$ by (3). Thus at least $h^2/5$ of the ordered pair mass in $X \times X$ is absent from B. All this missing mass lies in $N_A^+(i) \times N_R^+(i)$, so

$$M_1(i) \le a_i e_i - \frac{(a_i + e_i - 1)_+^2}{5}.$$

The same argument in A, applied to $N_R^-(i) \cap N_B^-(i)$, yields $M_3(i) \le K_5(d_i, f_i)$. Missing loops are included in the ordered pair mass, so no diagonal correction is needed. Summing the three fibre bounds proves (9). □

The next local inequality is the computer-assisted part of the proof. Its domain includes every six-tuple supplied by Lemma 3.1, regardless of the number of colours.

Lemma 3.2. *Let $a, b, c, d, e, f \in [0, 1]$, $x = \max(a, b) < 1$, and $y = \max(c, d) < 1$. Suppose that*

$$\min\{q_r, K_r(a, e)\} + bc + \min\{q_r, K_r(d, f)\} \ge 3p_r. \tag{10}$$

Define

$$\begin{aligned} F &= ae + bc + df - \frac{a + b + c + d + e + f}{2}, \\ G_r &= \alpha_r \left(\frac{1}{1 - x} + \frac{1}{1 - y} \right) + 4\alpha_r \left(\frac{1}{2 - y - e} + \frac{1}{2 - x - f} \right) - \gamma_r. \end{aligned} \tag{11}$$

Then

$$F \le G_r + \kappa_r(b - a + c - d). \tag{12}$$

Proof of Lemma 3.2. We first handle a neighbourhood of the equality configuration analytically. We then describe the finite certificates covering its remaining domain.

Set $u = (x + f)/2$, $v = (y + e)/2$, and

$$F_0 = xe + xy + yf - x - y - \frac{e + f}{2}, \qquad h_r = \frac{1}{2} + \frac{1}{r}.$$

Suppose that $x, y, u, v \ge h_r$. If $r = 6, 7$, also suppose that all six original variables are at least $1/2$. Increasing a, b to x and c, d to y does not decrease F, since the corresponding partial derivatives are $e - 1/2, c - 1/2, b - 1/2, f - 1/2$. Hence $F \le F_0$. If $r = 5$, instead suppose that the six variables are at least $3/5$. Direct expansion gives

$$\begin{aligned} F_0 - F + \tfrac{1}{10}(b - a + c - d) = (x - a)(e - \tfrac{2}{5}) + (x - b)(y - \tfrac{3}{5}) \\ + (y - c)(b - \tfrac{3}{5}) + (y - d)(f - \tfrac{2}{5}) \ge 0. \end{aligned}$$

Thus in both cases it suffices to prove $G_r \ge F_0$.

For $h_r \le t < 1$, the identity

$$\frac{\alpha_r}{1 - t} - \frac{t^2 - t}{2} - \frac{\gamma_r}{6} = \frac{(t - q_r)^2(t - 2/r)}{2(1 - t)} \ge \frac{(t - q_r)^2}{2}$$

holds because $(t-2/r)/(1-t) \geq 1$. In (11), the last two reciprocal terms become $2\alpha_r/(1-v)$ and $2\alpha_r/(1-u)$. Apply the displayed inequality once to x, y and twice to u, v. Writing $S = x+y$, $U = u+v$, $X = x-y$, and $V = u-v$, expansion gives

$$G_r - F_0 \geq \frac{(S-U)^2}{2} + \frac{(S-2q_r)^2}{4} + \frac{(U-2q_r)^2}{2} + \left(\frac{X}{2}+V\right)^2 \geq 0. \tag{13}$$

This proves (12) on the specified analytic region.

For the rest of the domain, parameterize the two maxima by

$$(a,b) = (x, xz) \text{ or } (xz, x), \qquad (c,d) = (y, yw) \text{ or } (yw, y),$$

with $(x, z, y, w, e, f) \in [0,1]^6$. These four choices cover all six-tuples, including $x = 0$ or $y = 0$. Let

$$(L_1, L_2, L_3, L_4) = (1-x, 1-y, 2-y-e, 2-x-f), \qquad D = L_1 L_2 L_3 L_4.$$

The denominators in Lemma 3.2 are positive, and multiplication of $G_r - F + \kappa_r(b-a+c-d)$ by $2r^3 D$ gives the polynomial

$$\begin{aligned} N_r = (r-2)\sum_{j=1}^{4} \omega_j \prod_{t \neq j} L_t - 6r(2r-3)D - 2r^3 FD \\ + 2r^3 \kappa_r D(b-a+c-d), \qquad (\omega_1, \omega_2, \omega_3, \omega_4) = (1,1,4,4). \end{aligned} \tag{14}$$

Its coordinate degrees are at most $(3,1,3,1,2,2)$. The coefficients are integers; in particular $2 \cdot 5^3 \kappa_5 = 25$.

For completeness, a tensor Bernstein representation of coordinate degrees $(d_1, \ldots, d_6)$ is an expansion in the basis

$$\prod_{j=1}^{6} \binom{d_j}{i_j} t_j^{i_j} (1-t_j)^{d_j - i_j}, \qquad 0 \leq i_j \leq d_j.$$

Every basis function is nonnegative on the unit cube, and their sum is one. Consequently, nonnegative Bernstein coefficients imply that the represented polynomial is nonnegative throughout the cube. The same conclusion holds on any axis-parallel box after an affine change of coordinates.

For each of the twelve choices of r and maximum branch, the supplement provides the initial Bernstein coefficients of (14), multiplied by a positive integer, and a complete binary subdivision tree. Each internal node bisects one coordinate interval. A leaf is justified in one of three ways. First, its upper corner may violate (10), so the whole box lies outside the domain. Second, all Bernstein coefficients of N_r on that box may be nonnegative. Third, its lower corner may satisfy the analytic conditions leading to (13).

The first justification uses monotonicity. For $r = 5$, the function K_5 is nondecreasing in both coordinates. Indeed, when $s + t > 1$,

$$\frac{\partial K_5}{\partial s} = t - \frac{2}{5}(s+t-1) \geq \frac{3}{5} t \geq 0,$$

and the other derivative is obtained by exchanging s, t. On $s + t \leq 1$, the claim follows from $K_5 = st$. For $r = 6, 7$ it is immediate from $K_r = st$. Each original variable is also nondecreasing in every branch parameter. The third justification uses the same monotonicity for the lower-corner conditions on the original variables and on x, y, u, v.

Here is how the verifier certifies the polynomial and every subdivision. It evaluates (14) and its initial Bernstein representation at the grid $t_j = i_j/d_j$. There are $4 \cdot 2 \cdot 4 \cdot 2 \cdot 3 \cdot 3 = 576$ grid points.

Equality at this full tensor grid, checked over the rationals, proves polynomial identity because of the coordinate degree bounds. It is not a test at a numerical mesh. On a bisection, the verifier recomputes the child coefficients by the de Casteljau recurrence. It multiplies both children by the positive factor 2^{d_j}, so all stored and recomputed coefficients remain integers. The paths must form a complete binary tree, with exactly the two halves of the same coordinate at each node. This verifies coverage as well as positivity.

Table 1 records the complete checks. The domain leaves discard boxes outside the hypothesis, the Bernstein leaves certify nonnegative coefficients, and the analytic leaves use the proof above. Together with the tree coverage, these tests establish the inequality on the whole continuous domain, including points that do not arise from a palette. The program reads no floating-point values and calls no optimization solver. This proves Lemma 3.2.

Table 1: Exact subdivision certificates for Lemma 3.2.

	$r = 5$	$r = 6, 7$ combined
Maximum branches	4	8
Tree nodes	591302	45390
Domain leaves	146632	10721
Bernstein leaves	148955	11896
Analytic leaves	66	82
Total leaves	295653	22699
Maximum depth	63	43

The certificate files are `supplement/six/certificates.json` and `supplement/seven-eight/certificates.json`. The adjacent `verify.py` files reconstruct the polynomial identities and the full trees. These twelve branches are checked by one implementation of the Bernstein procedure; a second independent implementation is supplied only for the pattern certificates in Section 4. Certificate hashes and complete verification summaries are included in the supplement. □

Proof of Theorem 1.1. Suppose some finite S_{r+1}-bad palette has density greater than p_r. Apply Lemma 3.1 to obtain a maximizing positive weight vector and its projection degrees. The six-tuple at every colour satisfies (10), so Lemma 3.2 applies. The correction term has zero weighted mean, since

$$\sum_i w_i(b_i - a_i) = 0, \qquad \sum_i w_i(c_i - d_i) = 0.$$

Average (12) over the colours and substitute it into (8). The three reciprocal bounds (6)–(7) yield

$$P^* \le 1 + 2\alpha_r r + 4\alpha_r r - \gamma_r = 1 - \frac{3}{r} + \frac{3}{r^2} = p_r,$$

contrary to $P^* > p_r$. Thus every finite bad palette, with every choice of weights, has density at most p_r. Equation (2) gives the uniform Turán upper bound. Lemma 2.1 attains it, proving the three asserted values. The conclusion for $k \ge 6$ follows on adjoining the theorem of Lin and Zhou for $k \ge 9$. □

4 Bounds for four and five leaves

For S_4 and S_5, we replace the six-variable estimate by an inequality for three independent colours. The auxiliary digraph from Section 2 supplies both the forbidden finite patterns and additional

neighbourhood inequalities. This keeps the number of variables fixed while allowing an arbitrary colour set and arbitrary weights.

Fix $k \in \{4, 5\}$ and an S_k-bad palette with law w. Let

$$p = \mathbb{E}[A_{IJ} B_{JK} R_{IK}], \tag{15}$$

where I, J, K are independent colours of law w. The original density satisfies $P(w) \leq p$, since every admissible triple belongs to all three projections. No equality between the palette and its projection closure is assumed.

Draw independent colours I_0, I_1, I_2. Pulling back the arcs of D gives a pattern H on $\{\mathsf{L}, \mathsf{R}\} \times \{0, 1, 2\}$, whose vertices we denote by $\mathsf{L}i$ and $\mathsf{R}i$. Even when sampled colours coincide, H is T_k-free. Any tournament in H would have distinct images in D, because two vertices mapping to the same vertex have no arc between them. It would therefore give a tournament in D, contradicting Lemma 2.2. We encode the pattern by twenty-one bits in the following order

$$A_{01}, A_{02}, A_{10}, A_{12}, A_{20}, A_{21}, \quad B_{01}, B_{02}, B_{10}, B_{12}, B_{20}, B_{21},$$
$$R_{00}, R_{01}, R_{02}, R_{10}, R_{11}, R_{12}, R_{20}, R_{21}, R_{22}.$$

Here $A_{ij} = A_{I_i I_j}$, and the other relations are interpreted in the same way. The diagonal entries of A and B are zero. The symbol $H_{ui,vj}$ denotes the arc indicator from side u, index i, to side v, index j.

Let $\sum_{(i,j,\ell)}$ denote a sum over the six permutations of $(0, 1, 2)$. Define

$$p(H) = \frac{1}{6} \sum_{(i,j,\ell)} A_{ij} B_{j\ell} R_{i\ell}.$$

Exchangeability gives $\mathbb{E}p(H) = p$. Distinct formal indices do not require distinct sampled colours. All later expectations retain these possible coincidences.

We next define two classes of quantities with nonnegative expectation. The first is a positive semidefinite form. For a root index i and an extension j, put

$$f_{ij} = (A_{ij}, A_{ji}, B_{ij}, B_{ji}, R_{ij}, R_{ji}, R_{jj}) \in \{0, 1\}^7.$$

For $f \in \{0, 1\}^7$, let $\Phi(f)$ be the vector of the 29 squarefree monomials in its bits of degrees zero, one, and two, ordered first by degree and then lexicographically. If Q_0, Q_1 are positive semidefinite 29×29 matrices, set

$$z(H) = \frac{1}{3} \sum_{i=0}^{2} \Phi(f_{ij})^{\mathsf{T}} Q_{R_{ii}} \Phi(f_{i\ell}), \qquad \{j, \ell\} = \{0, 1, 2\} \setminus \{i\}. \tag{16}$$

The order of j, ℓ is immaterial because the matrices are symmetric. Conditional on the actual root colour I_i, the two extensions are independent and identically distributed. If $v(I_i) = \mathbb{E}[\Phi(f_{ij}) \mid I_i]$, then the corresponding conditional expectation in (16) is

$$v(I_i)^{\mathsf{T}} Q_{R_{I_i I_i}} v(I_i) \geq 0.$$

It follows that $\mathbb{E}z(H) \geq 0$. In particular, no assertion that $z(H)$ is nonnegative for each individual pattern is needed.

The second class uses arc bounds inside neighbourhoods. Every in- or outneighbourhood in D is T_{k-1}-free. Define $\rho_k = (k-3)/(k-2)$. By (3), a nonnegative mass assignment of total z to such a neighbourhood has arc mass at most $\rho_k z^2$.

These masses may depend on the relations between the root colour and the extension colour. Define fifteen nonnegative selector functions on seven-bit flags by

$$\ell_0(f) = 1, \qquad \ell_{2a+1}(f) = f_a, \qquad \ell_{2a+2}(f) = 1 - f_a \quad (0 \le a \le 6).$$

A descriptor consists of a side $s \in \{\mathsf{L}, \mathsf{R}\}$, an orientation $\epsilon \in \{+, -\}$, an integer $h \in \{0, 1, 2, 3, 4\}$, a root type $b \in \{-1, 0, 1\}$, and a selector index $a \in \{0, \dots, 14\}$. Put $\omega_{\mathsf{L}} = h/4$, $\omega_{\mathsf{R}} = 1 - h/4$, and let $\chi_b(i)$ equal 1 when $b = -1$ and otherwise equal $\mathbf{1}[R_{ii} = b]$. The neighbourhood indicator is

$$n_{si,uj} = \begin{cases} H_{si,uj}, & \epsilon = +, \\ H_{uj,si}, & \epsilon = -. \end{cases}$$

For this descriptor, define

$$\begin{aligned} g(H) = \frac{1}{6} \sum_{(i,j,\ell)} & \chi_b(i) \ell_a(f_{ij}) \ell_a(f_{i\ell}) \\ & \cdot \sum_{u,v \in \{\mathsf{L},\mathsf{R}\}} \omega_u \omega_v n_{si,uj} n_{si,v\ell} (\rho_k - H_{uj,v\ell}). \end{aligned} \tag{17}$$

To see that $\mathbb{E}g(H) \ge 0$, condition on the root colour. On vertex (u, c) of the chosen neighbourhood, assign mass $\omega_u w_c \ell_a(f_{ic})$. The conditional expectation in (17) is exactly $\chi_b(i)(\rho_k z^2 - m)$, where z is the total assigned mass and m its directed arc mass. Lemma 2.3 gives its nonnegativity. The root is sampled with law w; it is not given an additional factor ω_s. All selector values depend only on the root and a single extension, so the independence used in this calculation is preserved.

The finite certificate below is a pointwise inequality on relation patterns. The nonnegative expectations proved above then turn it into a bound for every weighted palette. The finite enumeration verifies the rational inequality, while conditional independence and the neighbourhood bound justify averaging it over an arbitrary colour set.

Lemma 4.1. *For $k = 4, 5$, there exist rational positive definite matrices Q_0, Q_1 and nonnegative rational coefficients λ_ν for descriptors of the form above such that every T_k-free twenty-one-bit pattern satisfies*

$$p(H) + z(H) + \sum_{\nu} \lambda_\nu g_\nu(H) \le c_k, \qquad c_4 = \frac{37}{100}, \quad c_5 = \frac{12}{25}. \tag{18}$$

The numbers of descriptors are 110 *and* 32*, respectively.*

Proof. The matrices and descriptor lists are supplied in `supplement/four/certificate.json` and `supplement/five/certificate.json`. All matrix entries and multipliers have common denominator $D_0 = 10^8$. Each descriptor explicitly records h, the side, orientation, root type, selector, and multiplier. The matrices are symmetric. Exact rational elimination of each matrix gives 29 positive pivots, proving positive definiteness.

The verifier enumerates all 2^{21} assignments to the displayed bits. For each assignment, it constructs the corresponding six-vertex digraph and tests every injection of the ordered vertices of T_k. An injection is present precisely when all its forward arcs are present. There are 72 distinct nontrivial required-bit masks for each k after redundant copies of the same mask are removed. No orbit representative reduction is used. The surviving patterns and the exact minimum slack in (18) are listed in Table 2.

Table 2: Complete pattern verification for Lemma 4.1.

k	Allowed patterns	Descriptors	Minimum slack
4	1417266	110	5171600/19200000000
5	2049852	32	31684636/28800000000

We specify the arithmetic normalization used for the check. Let $Z_b = D_0 Q_b$ and let $m_\nu = D_0 \lambda_\nu$ be the stored integers. For a fixed context (h, s, ϵ) and two flags f, g, the verifier forms

$$T_b(f,g) = \sum_{\substack{\nu \text{ in this context} \\ b_\nu \in \{-1,b\}}} m_\nu \ell_{a_\nu}(f)\ell_{a_\nu}(g).$$

This merely collects the descriptors with the same side proportion and neighbourhood. For $k = 4$, a common denominator is $192D_0$. The numerator of the objective term is $32D_0$ times the six-permutation objective count; the Gram term has numerator $64\sum_i \Phi(f_{ij})^{\mathsf{T}} Z_{R_{ii}} \Phi(f_{i\ell})$. For a root i and the other indices j, ℓ, the numerator from one context is

$$2T_{R_{ii}}(f_{ij}, f_{i\ell}) \sum_{u,v} h_u h_v n_{si,uj} n_{si,v\ell}(1 - H_{uj,v\ell} - H_{v\ell,uj}),$$

where $(h_{\mathsf{L}}, h_{\mathsf{R}}) = (h, 4-h)$. This follows by pairing the two orders of the extensions in (17).

For $k = 5$, the common denominator is $288D_0$. The objective and Gram factors become $48D_0$ and 96, respectively. The context numerator becomes

$$T_{R_{ii}}(f_{ij}, f_{i\ell}) \sum_{u,v} h_u h_v n_{si,uj} n_{si,v\ell}(4 - 3H_{uj,v\ell} - 3H_{v\ell,uj}).$$

These integer expressions are evaluated on every allowed pattern and compared with the appropriate multiple of c_k. Table 2 gives the resulting positive minimum differences after division by the common denominator. The supplementary verifiers also check feature order, descriptor domains, signs of multipliers, and the matrix elimination. A second implementation sums the descriptors directly rather than using the collected tables. Both implementations give the same pattern counts and minimum slacks. This proves (18) exactly. □

Proof of Theorem 1.2. For an arbitrary weighted S_k-bad palette, the sampled pattern is T_k-free. Taking expectations in (18) and using $\mathbb{E}z(H) \geq 0$ and $\mathbb{E}g_\nu(H) \geq 0$ gives $p \leq c_k$. Equation (15) therefore bounds the original palette density by c_k. The palette characterization (2) proves the two upper bounds. Lemma 2.1, with $r = 3, 4$, proves the lower bounds. □

Theorems 1.1 and 1.2 leave only the four-leaf and five-leaf cases of the star-density conjecture unresolved. Their conjectured values remain 1/3 and 7/16. We state convenient rational upper bounds; the positive slacks in Table 2 permit small numerical improvements, but leave a gap to the conjectured values. The present certificates do not determine sharp bounds or their equality cases.

Computational material and data availability. All certificate data and exact verifiers are included in the supplement. GPT-6.1 (OpenAI) assisted with implementing the verification programs. Their entry point is `run_all.py`. Verification requires Python's standard library and, for the pattern checks, a C++17 compiler. No optimization solver is required.

Declaration of AI use. The author used GPT-6.1 to assist with exploring proof strategies, checking calculations, preparing verification programs, and improving the exposition. The author takes full responsibility for the content of this paper.

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